

HIV & Opportunistic Infections – Summary

Key Symptoms: Depends on stage of disease

- Fever, weight loss, mouth sores, myalgia, neuropathy, malaise, lymphadenopathy, liver & spleen enlargement, N/V, frequent infection, etc.

Most common causing-pathogens: HIV-1 (most common worldwide; main focus), HIV-2

Routes of Transmission: Sexually, perinatally, parenterally and via breastfeeding

Diagnosis: The 2 key values to measure in HIV are the viral load and CD4+ count. These two values can help determine the stage of disease and direct us to the best possible treatment option. E.g., a CD4+ count <200 will indicate risk of opportunistic infections such as *P. jirovecii*, etc.

Pharmacological Treatment: Should be started ASAP upon diagnosis. The sooner a patient starts ART, the better the outcome. Combination therapy is essential, both for increased efficacy and to prevent the development of resistance.

- Anti-retroviral therapy (ART) regimens **MUST** contain 2 NRTIs (nucleos(t)ide reverse-transcriptase inhibitors) and one of the following for a 3-drug regimen:
 - NNRTI (Non-NRTI) **OR** PI (protease inhibitor) + PK booster **OR** an INSTI (integrase inhibitor)
- ❖ **See slides 33-38 for prophylaxis and treatment against susceptible infections based on CD4+ count**

Monitoring Parameters: Viral load and CD4+ count → HIV diagnosis, acute retroviral syndrome, general therapy monitoring/modification

Additional Key Points

- Important to know drug/class specific contraindications
 - Abacavir should be avoided in HLA-B*5701 status positive or unknown patients
 - Efavirenz must be taken on an empty stomach and should be avoided with many comorbidities e.g., pregnancy, psychiatric illness, prolonged QT interval, dementia, dyslipidemia, etc.
- Entry and fusion inhibitors are not commonly used
- Adherence must be >95% (aim for 100%)
- If HBV co-infection, do NOT use dolutegravir/lamivudine, use tenofovir + (lamivudine **OR** emtricitabine) to prevent resistance
- BMD may decrease while on antiretroviral therapy, ensure adequate exercise, calcium & vitamin D intake

Key Interactions

- Be aware of multiple CYP450-related interactions – see slides 24-30
- Dolutegravir + Bictegravir both interfere with the renal elimination of metformin which increases risk of lactic acidosis
- Avoid taking antacids and multivitamins with integrase inhibitors because they decrease the absorption leading to reduced efficacy
- Tenofovir DF, when taken with NSAIDs (and other nephrotoxic drugs) increases the risk of acute kidney injury